

Qualification Pack



Solar PV Module Manufacturing Technician

QP Code: SGJ/Q0119

Version: 4.0

NSQF Level: 4

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Qualification Pack

Contents

SGJ/Q0119: Solar PV Module Manufacturing Technician	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SGJ/N0145: Carry out manufacturing of Solar PV Modules	5
SGJ/N0147: Maintain personal health & safety in a manufacturing facility	17
DGT/VSQ/N0102: Employability Skills (60 Hours)	21
Assessment Guidelines and Weightage	28
<i>Assessment Guidelines</i>	28
<i>Assessment Weightage</i>	29
Acronyms	30
Glossary	31

Qualification Pack

SGJ/Q0119: Solar PV Module Manufacturing Technician

Brief Job Description

Solar PV Module Manufacturing Technician checks the front glass cover for the PV module, monitors the process of soldering solar cells to the strings to make interconnect, lamination of modules, framing of solar PV module, module testing and packaging for transit.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it without any accidents so diligence and hardworking are desired attributes for individuals performing this role. He must also demonstrate strong work ethics, an ability to communicate courteously with co-workers and must be good with following instructions of the supervisor

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N0145: Carry out manufacturing of Solar PV Modules](#)
2. [SGJ/N0147: Maintain personal health & safety in a manufacturing facility](#)
3. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Assembly Line Operator
Country	India
NSQF Level	4
Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7422.1901

Qualification Pack

Minimum Educational Qualification & Experience	12th grade Pass (with Science or equivalent) with 1 Year of experience of relevant experience in renewable energy/power sector OR 10th grade pass with 3 Years of experience of relevant experience in renewable energy/power sector OR 10th grade pass with 2 Years of experience of any combination of NTC/NAC/CITS or equivalent in relevant trade OR Previous relevant Qualification of NSQF Level (3.5) OR Previous relevant Qualification of NSQF Level (3 - Solar Manufacturing-Junior Technician) with 3 Years of experience of relevant experience in renewable energy/power sector
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/05/2027
NSQC Approval Date	30/05/2024
Version	4.0
Reference code on NQR	QG-04-ES-02633-2024-V2-SCGJ
NQR Version	2

Remarks:

Total 420 hours i.e. 14 Credits (Theory : 180 + Practical : 120 + OJT : 120 Hours)

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SGJ/N0145: Carry out manufacturing of Solar PV Modules

Description

This unit is about manufacturing of Solar PV Modules which provides a broader overview of Solar PV Technology, Bill of Material of Solar Panel, soldering technology in module manufacturing, Solar Module Manufacturing process, Visual inspection and Electroluminescence (EL) Test as well as Packing of Solar PV Modules

Scope

The scope covers the following :

- Role and Responsibilities of a Manufacturing Technician
- BOM of Solar Module
- Soldering process
- Manufacturing process
- Testing of PV Module
- Packing of Solar Module

Elements and Performance Criteria

Introduction to Solar PV Technology

To be competent, the user/individual on the job must be able to:

- PC1.** identify the role of Solar Photovoltaic (PV) Manufacturing technician, its importance in the sector and the advantages of doing the course.
- PC2.** show how irradiation influence performance of solar module.
- PC3.** show how to the use of Hand-held irradiation meter.
- PC4.** show Solar Module Construction & Manufacturing Process flow Chart.
- PC5.** identify mono and poly crystalline silicon solar cell technology and recent trends.
- PC6.** identify the concept of solar energy, working of solar cell, Electrical Properties of Solar Cell, Solar PV Applications
- PC7.** identify the property of semiconductor material and P-N Junction.
- PC8.** discuss the basic aspects of solar energy and power generation.
- PC9.** definition of properties of light, global horizontal Irradiation (GHI) and Direct Normal irradiance.
- PC10.** discuss the various configurations of Solar PV Power plants. (Grid connected, Off grid etc).
- PC11.** show how to interpret signs, notices and/or cautions at project site.

Prepare and identify the BOM and its specification for solar module

To be competent, the user/individual on the job must be able to:

- PC12.** show each component and explain their application in Solar module
- PC13.** show how to prepare the BOM of Solar Module.
- PC14.** Identify various Materials used in solar module manufacturing as Solar Glass, Cell, EVA Back sheet, Aluminium Frame, Junction Box, Interconnect Ribbon, Silicon Sealant

Qualification Pack

- PC15.** show how to use AI based tool to analyse performance of module.
- PC16.** discuss technical specifications and their role /importance in module performance & reliability.

Use of soldering technology in module manufacturing test

To be competent, the user/individual on the job must be able to:

- PC17.** discuss soldering process general requirement.
- PC18.** discuss tools used for Soldering process.
- PC19.** discuss Soldering Tests & Defect Identification.
- PC20.** discuss how to identification of defects (Dry soldering, Poor Soldering) & their criticality in PV module Reliability.
- PC21.** discuss how to Avoid wrong practices during Soldering Process.
- PC22.** discuss soldering process of Solar Cell String Bussing – interconnect Copper Ribbon.
- PC23.** discuss the Junction Box Soldering techniques & desoldering process,
- PC24.** discuss how to perform cell soldering and soldering of cell strings by string.
- PC25.** show how to perform soldering.
- PC26.** show how to the Junction box soldering techniques with hands on experience.
- PC27.** show how to use of soldering tools during soldering of solar cell.
- PC28.** show how to do plain soldering, soldering in junction boxes, soldering on cells to achieve interconnection.

Solar Module Manufacturing process (Solar Cell Tabbing/Ribbon cutting)

To be competent, the user/individual on the job must be able to:

- PC29.** identify the flow chart of Solar module manufacturing processes
- PC30.** discuss use of laser cutting machine, for cells cutting.
- PC31.** discuss the functioning of Solar cell Tabber Stringer machine for tabbing and interconnection of solar cell in to string by soldering them together.
- PC32.** discuss tabbing of cells interconnect ribbon on solar cells.
- PC33.** discuss how to removal of defective broken cells from string.
- PC34.** discuss solar cells string bussing interconnect of copper ribbon.
- PC35.** discuss the process of bussing and bus bar welding.
- PC36.** discuss how to monitor the assembly and interconnection of cells with metal ribbons to make a module.
- PC37.** discuss the process of EVA and back sheet layup.
- PC38.** discuss the process of lamination and its importance solar module performance.
- PC39.** discuss how to ensure there should not be voids, bubbles, back sheet wrinkles, and other defects during the visual inspections after the lamination process.
- PC40.** discuss the process of aluminium gluing /manipulating and framing process
- PC41.** show how to solar cell tabbing technique.
- PC42.** show how to functioning of Solar cell Tabber Stringer machine for tabbing and interconnection of solar cell in to string.
- PC43.** show how to perform Sorting of the solar cells using a cell sorter and put use the cells with same specifications for making a cell string.
- PC44.** show how to cut EVA using a cutter and place it on the glass substrate.

Qualification Pack

- PC45.** show how to perform monitoring the process of aligning and placing cell strings on EVA sheet at assembler module layup station.
- PC46.** show how to monitor the process of completing the module laminate circuit at the busing station and provide output leads.
- PC47.** show how to Monitor the process to laminate the assembly and cure the EVA with the laminator.
- PC48.** show how to Trim the edges of the laminated module panel using cutter.

Visual inspection and electroluminescence (EL) test

To be competent, the user/individual on the job must be able to:

- PC49.** discuss how to perform EL test of solar module and identify any dead and low power cells, short circuit cells, cracks, etc. If any such error is spotted, the module is sent back for fixing the error.
- PC50.** discuss visual inspection to identify defects, consequences of Soldering Process & It's Severity.
- PC51.** discuss the important checkpoints of string & laminate.
- PC52.** discuss how to check its output current, voltage and power of solar module.
- PC53.** discuss use of sun simulator for performing IV test characteristics of solar module.
- PC54.** discuss the flash test and measures the open-circuit voltage (VOC), voltage at maximum power point (VMP), short-circuit current (ISC), current at maximum power point (Imp), maximum power (Pm (W)) and the fill factor (FF). The parameters of the modules are measured at standard test conditions (STC).
- PC55.** discuss how to generate a Flash test report (FTR) for each module's output data.
- PC56.** discuss the module is connected to check its output current, voltage, power, etc
- PC57.** show how to solar cell tabbing technique.
- PC58.** show how to operate sun simulator and perform the various test.
- PC59.** show how to perform IV/ EL/Insulation test.
- PC60.** show how to perform the visual Inspection.
- PC61.** show how to generate flash test report.

Packing of Solar PV modules

To be competent, the user/individual on the job must be able to:

- PC62.** discuss how to visually inspect the completed module for quality of materials and workmanship.
- PC63.** discuss how to check proper packaging material for module.
- PC64.** discuss how to pack modules in properly designed cartons for transportation.
- PC65.** discuss how to keep module in to stacked in a box.
- PC66.** discuss the use of carton separators and also many available plastic separators.
- PC67.** identify processes where material and resources utilization can be optimized.
- PC68.** show how to the Automatic Packing process and handling of Solar Module.
- PC69.** show how to use of fork lift machine for moving of solar module pallet from one place to other.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** government/corporate policies and guidelines on: workplace safety, identification and mitigation of safety hazards, work procedures and guidelines for working at height
- KU2.** the process to document information using appropriate forms
- KU3.** the process of authorization from specified field safety officer and supervisor
- KU4.** complete knowhow on manufacturers warranty policy
- KU5.** basics of electrical concepts like voltage, current, power, energy, etc.
- KU6.** basics of electrical measurement equipments
- KU7.** basics on solar energy system and power generation
- KU8.** basics of electricity and associated terminologies like volts, amps and watts
- KU9.** basics of the electrical systems, components like strings, junction boxes, fuses, inverters, auxiliary power supplies, etc, their functioning and maintenance
- KU10.** use and maintenance of various measuring equipments such as multi-meter, solar array tester and meggers etc.
- KU11.** raw materials, shelf life, storage specifications of various equipment
- KU12.** cutting of EVA and back sheet
- KU13.** performance characteristic parameters of solar cells and modules
- KU14.** soldering techniques
- KU15.** usage and handling procedures of solar panels
- KU16.** operation of laminator, framing machine, etc.
- KU17.** various types of faults that can occur in the electrical systems/ components, their detection equipment and methods
- KU18.** various types of tools and procedures involved in maintenance and troubleshooting the system
- KU19.** various types of tools, measuring equipments and procedures involved in maintenance and troubleshooting of electrical components of the solar module manufacturing plants
- KU20.** importance of wearing protective clothing and other safety gear while carrying out maintenance activities (PPEs)
- KU21.** proper documentation as per standards
- KU22.** relevant occupational health and safety standards importance of wearing protective clothing and other safety gear while carrying out manufacturing activities (PPEs)
- KU23.** organizational procedures for minimizing waste
- KU24.** efficient and inefficient utilization of material and water

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** undertake proper documentation as per relevant industry standards
- GS2.** clearly read and write vernacular/English language
- GS3.** read manuals, health and safety instructions, memos, other company documents
- GS4.** read from different sources- books, screens in machines and signage

Qualification Pack

- GS5.** the various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with peers, superiors and sub-ordinates
- GS10.** follow organisation rule- based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize work schedule to meet deadlines
- GS13.** work constructively and collaboratively with others
- GS14.** follow organisation code of conduct
- GS15.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS16.** recognize problems and search for solutions
- GS17.** choose best methods to complete assigned tasks
- GS18.** approach relevant authority when required
- GS19.** apply domain knowledge, observations and data to select course of action to perform tasks related to solar photovoltaic module manufacturing facility
- GS20.** use reasoning skills to identify and resolve basic problems
- GS21.** use intuition to detect any potential problems which could arise during operations
- GS22.** use acquired knowledge of the process for identifying and handling issues

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Solar PV Technology</i>	14	8	-	-
PC1. identify the role of Solar Photovoltaic (PV) Manufacturing technician, its importance in the sector and the advantages of doing the course.	2	-	-	-
PC2. show how irradiation influence performance of solar module.	-	2	-	-
PC3. show how to the use of Hand-held irradiation meter.	-	2	-	-
PC4. show Solar Module Construction & Manufacturing Process flow Chart.	-	2	-	-
PC5. identify mono and poly crystalline silicon solar cell technology and recent trends.	2	-	-	-
PC6. identify the concept of solar energy, working of solar cell, Electrical Properties of Solar Cell, Solar PV Applications	2	-	-	-
PC7. identify the property of semiconductor material and P-N Junction.	2	-	-	-
PC8. discuss the basic aspects of solar energy and power generation.	2	-	-	-
PC9. definition of properties of light, global horizontal Irradiation (GHI) and Direct Normal irradiance.	2	-	-	-
PC10. discuss the various configurations of Solar PV Power plants. (Grid connected, Off grid etc).	2	-	-	-
PC11. show how to interpret signs, notices and/or cautions at project site.	-	2	-	-
<i>Prepare and identify the BOM and its specification for solar module</i>	4	6	-	-
PC12. show each component and explain their application in Solar module	-	2	-	-
PC13. show how to prepare the BOM of Solar Module.	-	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Identify various Materials used in solar module manufacturing as Solar Glass, Cell, EVA Back sheet, Aluminium Frame, Junction Box, Interconnect Ribbon, Silicon Sealant	2	-	-	-
PC15. show how to use AI based tool to analyse performance of module.	-	2	-	-
PC16. discuss technical specifications and their role /importance in module performance & reliability.	2	-	-	-
<i>Use of soldering technology in module manufacturing test</i>	16	9	-	-
PC17. discuss soldering process general requirement.	2	-	-	-
PC18. discuss tools used for Soldering process.	2	-	-	-
PC19. discuss Soldering Tests & Defect Identification.	2	-	-	-
PC20. discuss how to identification of defects (Dry soldering, Poor Soldering) & their criticality in PV module Reliability.	2	-	-	-
PC21. discuss how to Avoid wrong practices during Soldering Process.	2	-	-	-
PC22. discuss soldering process of Solar Cell String Bussing – interconnect Copper Ribbon.	2	-	-	-
PC23. discuss the Junction Box Soldering techniques & desoldering process,	2	-	-	-
PC24. discuss how to perform cell soldering and soldering of cell strings by string.	2	-	-	-
PC25. show how to perform soldering.	-	2	-	-
PC26. show how to the Junction box soldering techniques with hands on experience.	-	2	-	-
PC27. show how to use of soldering tools during soldering of solar cell.	-	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. show how to do plain soldering, soldering in junction boxes, soldering on cells to achieve interconnection.	-	3	-	-
<i>Solar Module Manufacturing process (Solar Cell Tabbing/Ribbon cutting)</i>	25	24	-	-
PC29. identify the flow chart of Solar module manufacturing processes	2	-	-	-
PC30. discuss use of laser cutting machine, for cells cutting.	2	-	-	-
PC31. discuss the functioning of Solar cell Tabber Stringer machine for tabbing and interconnection of solar cell in to string by soldering them together.	3	-	-	-
PC32. discuss tabbing of cells interconnect ribbon on solar cells.	2	-	-	-
PC33. discuss how to removal of defective broken cells from string.	2	-	-	-
PC34. discuss solar cells string bussing interconnect of copper ribbon.	2	-	-	-
PC35. discuss the process of bussing and bus bar welding.	2	-	-	-
PC36. discuss how to monitor the assembly and interconnection of cells with metal ribbons to make a module.	2	-	-	-
PC37. discuss the process of EVA and back sheet layup.	2	-	-	-
PC38. discuss the process of lamination and its importance solar module performance.	2	-	-	-
PC39. discuss how to ensure there should not be voids, bubbles, back sheet wrinkles, and other defects during the visual inspections after the lamination process.	2	-	-	-
PC40. discuss the process of aluminium gluing /manipulating and framing process	2	-	-	-
PC41. show how to solar cell tabbing technique.	-	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC42. show how to functioning of Solar cell Tabber Stringer machine for tabbing and interconnection of solar cell in to string.	-	3	-	-
PC43. show how to perform Sorting of the solar cells using a cell sorter and put use the cells with same specifications for making a cell string.	-	3	-	-
PC44. show how to cut EVA using a cutter and place it on the glass substrate.	-	3	-	-
PC45. show how to perform monitoring the process of aligning and placing cell strings on EVA sheet at assembler module layup station.	-	3	-	-
PC46. show how to monitor the process of completing the module laminate circuit at the busing station and provide output leads.	-	3	-	-
PC47. show how to Monitor the process to laminate the assembly and cure the EVA with the laminator.	-	3	-	-
PC48. show how to Trim the edges of the laminated module penal using cutter.	-	3	-	-
<i>Visual inspection and electroluminescence (EL) test</i>	17	11	-	-
PC49. discuss how to perform EL test of solar module and identify any dead and low power cells, short circuit cells, cracks, etc. If any such error is spotted, the module is sent back for fixing the error.	3	-	-	-
PC50. discuss visual inspection to identify defects, consequences of Soldering Process & It's Severity.	2	-	-	-
PC51. discuss the important checkpoints of string & laminate.	2	-	-	-
PC52. discuss how to check its output current, voltage and power of solar module.	2	-	-	-
PC53. discuss use of sun simulator for performing IV test characteristics of solar module.	2	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC54. discuss the flash test and measures the open-circuit voltage (VOC), voltage at maximum power point (VMP), short- circuit current (ISC), current at maximum power point (Imp), maximum power (Pm (W)) and the fill factor (FF). The parameters of the modules are measured at standard test conditions (STC).	2	-	-	-
PC55. discuss how to generate a Flash test report (FTR) for each module's output data.	2	-	-	-
PC56. discuss the module is connected to check its output current, voltage, power, etc	2	-	-	-
PC57. show how to solar cell tabbing technique.	-	3	-	-
PC58. show how to operate sun simulator and perform the various test.	-	2	-	-
PC59. show how to perform IV/ EL/Insulation test.	-	2	-	-
PC60. show how to perform the visual Inspection.	-	2	-	-
PC61. show how to generate flash test report.	-	2	-	-
<i>Packing of Solar PV modules</i>	12	4	-	-
PC62. discuss how to visually inspect the completed module for quality of materials and workmanship.	2	-	-	-
PC63. discuss how to check proper packaging material for module.	2	-	-	-
PC64. discuss how to pack modules in properly designed cartons for transportation.	2	-	-	-
PC65. discuss how to keep module in to stacked in a box.	2	-	-	-
PC66. discuss the use of carton separators and also many available plastic separators.	2	-	-	-
PC67. identify processes where material and resources utilization can be optimized.	2	-	-	-
PC68. show how to the Automatic Packing process and handling of Solar Module.	-	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC69. show how to use of fork lift machine for moving of solar module pallet from one place to other.	-	2	-	-
NOS Total	88	62	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0145
NOS Name	Carry out manufacturing of Solar PV Modules
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Manufacturing
NSQF Level	4
Credits	7
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

SGJ/N0147: Maintain personal health & safety in a manufacturing facility

Description

This unit is about maintaining health & safety in a manufacturing facility by following established safe work procedure, maintaining personal protective equipment as well as proper use of required tools and equipment

Scope

The scope covers the following :

- Establish and Follow safe work procedure
- Use and maintain personal protective equipment
- Identify and mitigate safety hazards
- Demonstrate safe and proper use of required tools and equipment

Elements and Performance Criteria

Maintain personal health & safety in a manufacturing facility

To be competent, the user/individual on the job must be able to:

- PC1.** identify the requirements for safe work area.
- PC2.** discuss how to Administer first aid.
- PC3.** discuss how to identify the personal protective equipment used for the specific purpose.
- PC4.** discuss how to identify the electrical and chemical hazards associated with photovoltaic installations.
- PC5.** discuss how to identify work safety procedures and instructions to Handle Sharp and heavy component in a solar PV manufacturing.
- PC6.** discuss occupational health & safety standards and regulations for installation of Solar PV system.
- PC7.** show how to Administer first aid.
- PC8.** show how to Identify the electrical and chemical hazards associated with photovoltaic installations.
- PC9.** show how to Identify and perform work safety procedures and instructions to handle sharp and heavy component in a solar PV Manufacturing.
- PC10.** show how to good housekeeping practices and infection control guidelines.
- PC11.** show how to Dispose- off any waste materials in accordance with safe working practices and procedures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's installation policy
- KU2.** company's customer support policy

Qualification Pack

- KU3.** document information using appropriate corporate forms
- KU4.** obtain authorization from specified field safety officer and supervisor
- KU5.** relevant personal protective equipments required within the manufacturing facility
- KU6.** relevant standards and regulations to be followed in the manufacturing facility
- KU7.** occupational health and safety (OHS) standards to be followed in the manufacturing facility
- KU8.** risk identification and mitigation procedure for safe work in a manufacturing facility
- KU9.** knowhow of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** read vernacular/english language
- GS3.** read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** ability to read from different sources- books, screens in machines and signage
- GS5.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with peers, supervisor and sub-ordinates
- GS10.** follow organisation rule- based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize work schedule to meet deadlines
- GS13.** plan to utilise time and equipment's effectively
- GS14.** work constructively and collaboratively with others
- GS15.** follow organisational code of conduct
- GS16.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS17.** recognize problems and search for solutions
- GS18.** choose best methods to complete assigned tasks
- GS19.** approach relevant authority when required
- GS20.** apply domain knowledge, observations and data to select course of action to perform tasks related to the job role
- GS21.** critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities
- GS22.** ask questions for better understanding

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health & safety in a manufacturing facility</i>	25	25	-	-
PC1. identify the requirements for safe work area.	3	-	-	-
PC2. discuss how to Administer first aid.	4	-	-	-
PC3. discuss how to identify the personal protective equipment used for the specific purpose.	4	-	-	-
PC4. discuss how to identify the electrical and chemical hazards associated with photovoltaic installations.	4	-	-	-
PC5. discuss how to identify work safety procedures and instructions to Handle Sharp and heavy component in a solar PV manufacturing.	5	-	-	-
PC6. discuss occupational health & safety standards and regulations for installation of Solar PV system.	5	-	-	-
PC7. show how to Administer first aid.	-	5	-	-
PC8. show how to Identify the electrical and chemical hazards associated with photovoltaic installations.	-	5	-	-
PC9. show how to Identify and perform work safety procedures and instructions to handle sharp and heavy component in a solar PV Manufacturing.	-	5	-	-
PC10. show how to good housekeeping practices and infection control guidelines.	-	5	-	-
PC11. show how to Dispose- off any waste materials in accordance with safe working practices and procedures.	-	5	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0147
NOS Name	Maintain personal health & safety in a manufacturing facility
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health & Safety
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0145.Carry out manufacturing of Solar PV Modules	88	62	0	0	150	60
SGJ/N0147.Maintain personal health & safety in a manufacturing facility	25	25	0	0	50	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	20
Total	133	117	-	-	250	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.